SIMPLEX MEMORY NETWORKS: A GEOMETRIC FEEDFORWARD ARCHITECTURE FOR COUPLED SHORT- AND LONG-TERM MEMORY

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ABSTRACT. We introduce a simplex-based feedforward neural architecture built on a discrete lattice inside a regular n-simplex. The hidden nodes are the lattice points of the simplex, the input and output interfaces are realized by two opposite facets, and information propagates along a directed nearest-neighbor graph induced by a linear potential. The resulting architecture is a directed acyclic graph with a distinguished narrow backbone and a broad shared interface between the input and output facets. In the minimal cases of the triangle and the tetrahedron, the architecture reduces to MLP-like motifs with geometrically induced parallel channels and skip paths. We propose a geometric interpretation in which the backbone models long-term memory while the shared intermediate face models a broad short-term memory buffer. We conclude by outlining several mathematical directions, including architectural scaling in (n, m) and continuous-time or PDE limits on simplicial domains.

CONTENTS

1. INTRODUCTION

Standard feedforward neural networks organize computation along a linear sequence of layers, and their training is classically formulated through backpropagation [12]. Modern deep architectures often enrich this linear organization by introducing skip connections, residual pathways, or other mechanisms that improve information propagation across depth [9]. At the

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same time, memory-oriented models suggest that useful architectures should support both a stable, slowly integrated pathway and a rapidly accessible pathway for short-lived information [10, 13, 7].

A different but complementary line of work has emphasized that graphs alone encode only pairwise interactions, whereas simplicial and more general topological structures provide a natural framework for higher-order interactions [1, 5, 2, 8]. Most of this literature focuses on extending signal processing or message passing to simplicial domains. In the present note, our goal is different: we propose a *simplex-based feedforward memory architecture* whose geometry itself organizes input, output, hidden propagation, and memory coupling.

The basic idea is simple. We begin with a regular n-simplex, discretize it by a lattice of points, and view these lattice points as hidden nodes. Two opposite facets of the simplex act as input and output interfaces, while their common $(n-2)$-dimensional face serves as a shared intermediate buffer. A linear potential then induces a directed nearest-neighbor graph on the lattice, yielding a feedforward hidden architecture. In the minimal motifs, the triangle and the tetrahedron already exhibit the qualitative features we seek: a narrow backbone, parallel intermediate channels, and a natural distinction between deep and shallow propagation paths.

The contribution of this note is conceptual and mathematical. First, we define a family of directed hidden graphs $\mathrm{SMN}(n, m)$ based on the lattice of a regular n-simplex with m points on each edge. Second, we identify the input facet, output facet, and shared interface, and compute several basic combinatorial quantities associated with these sets. Third, we describe the triangle and tetrahedron motifs explicitly as minimal geometric instances of the construction. Finally, we propose a geometric interpretation in which a narrow deep backbone represents long-term memory, while a broad shallow interface represents short-term memory.

2. Geometric and Combinatorial Preliminaries

We begin by introducing the geometric foundation of the simplex memory architecture. The construction proceeds in two steps: first we define the continuous regular simplex, then we discretize it by a lattice of points that will serve as the hidden nodes of the network.

Definition 2.1 (The regular n-simplex)**.** For $n \geq 2$, the standard regular n-simplex is the subset of $\mathbb{R}^{n+1}$ defined by

$$\Delta^n := \left\{ x = (x_0, \ldots, x_n) \in \mathbb{R}^{n+1} : x_i \geq 0,\ \sum_{i=0}^{n} x_i = 1 \right\}.$$

This is an n-dimensional polytope lying in the affine hyperplane $\sum_{i=0}^{n} x_i = 1$. Its vertices are the canonical basis vectors

$$e_0, e_1, \ldots, e_n \in \mathbb{R}^{n+1}.$$

Throughout this work we distinguish two vertices and relabel them as

$$a := e_0, \qquad b := e_1, \qquad c_1 := e_2, \ \ldots, \ c_{n-1} := e_n.$$

Thus a and b play the roles of the two extreme vertices (the "backbone" endpoints), while $c_1, \ldots, c_{n-1}$ form the intermediate vertex set.

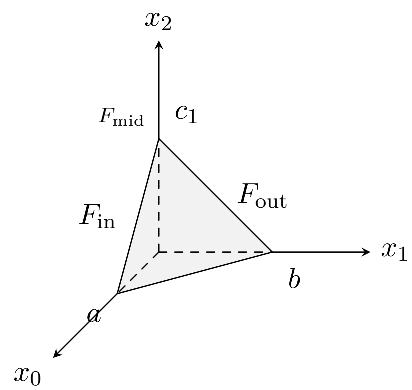

FIGURE 1. The regular 2-simplex $\Delta^2 \subset \mathbb{R}^3$, with vertices $a = e_0$, $b = e_1$, $c_1 = e_2$ on the coordinate axes. The input facet is $F_{\text{in}} = \operatorname{conv}(a, c_1)$, the output facet $F_{\text{out}} = \operatorname{conv}(c_1, b)$, and their intersection $F_{\text{mid}} = \{c_1\}$.

The geometric structure becomes a neural architecture only after discretization. We now introduce the lattice that provides the discrete set of hidden nodes.

Definition 2.2 (The discrete simplex lattice)**.** Fix an integer $m \geq 2$ and write $s := m - 1$. The lattice of hidden nodes is defined by

$$V_{n,m} := \left\{ \alpha = (\alpha_0, \ldots, \alpha_n) \in \mathbb{Z}_{\geq 0}^{n+1} : \sum_{i=0}^{n} \alpha_i = s \right\}.$$

Each $\alpha \in V_{n,m}$ corresponds to the geometric point

$$x(\alpha) = \frac{1}{s}\alpha \in \Delta^n.$$

Equivalently, $V_{n,m}$ is the set of discrete barycentric coordinates of the simplex lattice with exactly m points (including endpoints) on every one-dimensional edge.

The input and output interfaces of the architecture are realized by two opposite facets of the simplex. Their intersection forms a shared intermediate buffer.

Definition 2.3 (Input, output, and shared facets)**.** We define the input facet, output facet, and shared intermediate face by

$$F_{\text{in}} := \{\alpha \in V_{n,m} : \alpha_1 = 0\}, \qquad F_{\text{out}} := \{\alpha \in V_{n,m} : \alpha_0 = 0\},$$

$$F_{\mathrm{mid}} := F_{\mathrm{in}} \cap F_{\mathrm{out}} = \{\alpha \in V_{n,m} : \alpha_0 = \alpha_1 = 0\}.$$

Geometrically, these correspond to the facets

$$\mathrm{conv}(a, c_1, \ldots, c_{n-1}), \qquad \mathrm{conv}(c_1, \ldots, c_{n-1}, b),$$

and their common face

$$\mathrm{conv}(c_1, \ldots, c_{n-1}).$$

We now compute the cardinalities of these sets. These quantities determine the number of hidden nodes, the size of the input and output interfaces, and the scaling of the architecture with respect to the parameters (n, m).

Proposition 2.4 (Basic cardinalities)**.** *The lattice sizes are given by*

$$|V_{n,m}| = \binom{m+n-1}{n}, \qquad |F_{\mathrm{in}}| = |F_{\mathrm{out}}| = \binom{m+n-2}{n-1},$$

$$|F_{\mathrm{mid}}| = \binom{m+n-3}{n-2}.$$

Proof. All three formulas follow from standard stars-and-bars counting. The total number of hidden nodes equals the number of nonnegative integer solutions of

$$\alpha_0 + \cdots + \alpha_n = m - 1,$$

which is $\binom{m+n-1}{n}$. For the input facet, we impose $\alpha_1 = 0$, leaving

$$\alpha_0 + \alpha_2 + \cdots + \alpha_n = m - 1,$$

with n variables; hence $|F_{\mathrm{in}}| = \binom{m+n-2}{n-1}$. The output facet is identical by symmetry. Finally, the shared face is obtained by imposing both $\alpha_0 = 0$ and $\alpha_1 = 0$, leaving $n - 1$ variables and thus

$$|F_{\mathrm{mid}}| = \binom{m+n-3}{n-2}.$$

□

Remark 2.5 (Interface-to-volume ratio)**.** The proportion of lattice nodes lying on the input facet (and likewise on the output facet) is

$$\frac{|F_{\mathrm{in}}|}{|V_{n,m}|} = \frac{\binom{m+n-2}{n-1}}{\binom{m+n-1}{n}} = \frac{n}{n+m-1}.$$

This ratio suggests a structural interpretation. Increasing m deepens the simplex lattice while reducing the facet proportion, whereas increasing n enlarges the relative size of the input and output interfaces. In this sense, the pair (n, m) naturally controls the balance between broad interfaces and deep interior propagation.

3. Directed Simplex Memory Graph

The lattice $V_{n,m}$ defined in section 2 provides only the set of hidden nodes. To obtain a functional architecture, we must specify how these nodes are connected and how information flows through the network. This section constructs the directed hidden graph that defines the simplex memory network.

The construction proceeds in two steps. First, we connect lattice points that are nearest neighbors, yielding an undirected graph that captures the local geometry of the simplex. Second, we orient these edges using a linear potential function that encodes the desired direction of information flow—from the input facet toward the output facet. The resulting directed graph is acyclic and exhibits a natural layer structure with geometrically induced skip connections.

We begin with the undirected connectivity.

Definition 3.1 (Nearest neighbors)**.** Two nodes $\alpha, \alpha' \in V_{n,m}$ are nearest neighbors if and only if

$$\alpha' = \alpha + e_i - e_j$$

for some $i \neq j$ with $\alpha_j \geq 1$. Equivalently, α' is obtained from α by moving one unit of barycentric mass from the jth coordinate to the ith coordinate.

All such edges have the same Euclidean length in the ambient simplex:

$$\|x(\alpha') - x(\alpha)\| = \frac{1}{m-1}\|e_i - e_j\| = \frac{\sqrt{2}}{m-1}.$$

Thus the lattice induces a regular grid in which every node has a uniform local neighborhood structure.

However, the undirected graph alone does not specify a direction of information flow. To obtain a feedforward architecture, we must orient these edges. The orientation is determined by assigning potentials to the vertices of the simplex and extending linearly to the lattice.

Definition 3.2 (Vertex potentials and induced orientation)**.** Assign the vertex potentials

$$\beta_a = 0, \qquad \beta_b = 2, \qquad \beta_{c_1} = \cdots = \beta_{c_{n-1}} = 1.$$

Equivalently, $\beta_0 = 0$, $\beta_1 = 2$, and $\beta_2 = \cdots = \beta_n = 1$. The hidden potential $H : V_{n,m} \to \mathbb{R}$ is defined by

$$H(\alpha) := \sum_{k=0}^{n} \beta_k \alpha_k = 2\alpha_1 + \alpha_2 + \cdots + \alpha_n.$$

A directed edge is retained from α to $\alpha' = \alpha + e_i - e_j$ if and only if

$$\alpha_j \geq 1 \qquad \text{and} \qquad \beta_i > \beta_j.$$

Edges with $\beta_i = \beta_j$ are deleted.

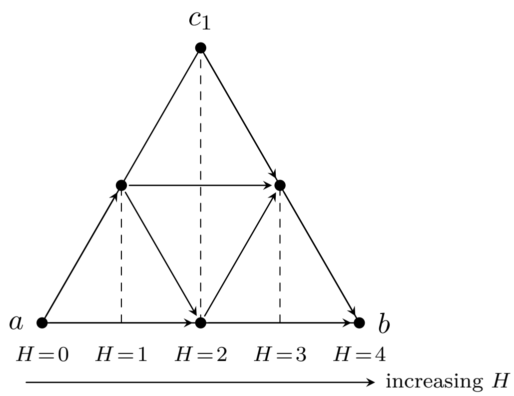

FIGURE 2. Potential flow on $\mathrm{SMN}(2,3)$ ($n = 2$, $m = 3$). Each node $\alpha \in V_{2,3}$ is shown at its geometric position in the equilateral triangle; its potential $H(\alpha) = 2\alpha_1 + \alpha_2$ equals its horizontal coordinate. Vertical dashed lines are isopotential sets. Solid arrows are $\Delta H = 1$ edges; bent arrows are $\Delta H = 2$ skip connections. All edges point strictly rightward.

This orientation rule has a simple interpretation: information flows from lower-potential vertices to higher-potential vertices. The admissible edge types are therefore exactly

$$a \to c_k, \qquad c_k \to b, \qquad a \to b, \qquad k = 1, \dots, n-1,$$

while all edges among the intermediate vertices $c_1, \dots, c_{n-1}$ are removed. Geometrically, this reflects the flow from the input vertex a (potential 0) through the intermediate layer (potential 1) to the output vertex b (potential 2), together with a direct skip connection from a to b.

The potential-based orientation immediately implies acyclicity.

Proposition 3.3 (Acyclicity)**.** *The directed hidden graph* $(V_{n,m}, E^{\to}_{n,m})$ *is acyclic.*

Proof. If $\alpha' = \alpha + e_i - e_j$ is a retained directed edge, then $H(\alpha') - H(\alpha) = \beta_i - \beta_j > 0$. Hence the potential strictly increases along every directed edge. A directed cycle would force the potential to strictly increase and return to its initial value, which is impossible. □

We can now give the formal definition of the architecture.

Definition 3.4 (Simplex memory network)**.** For integers $n \geq 2$ and $m \geq 2$, the *simplex memory network* $\mathrm{SMN}(n,m)$ is the directed hidden graph together with its distinguished input and output interfaces:

$$\mathrm{SMN}(n,m) := \left(V_{n,m}, E^{\to}_{n,m}, F_{\mathrm{in}}, F_{\mathrm{out}}\right).$$

Having defined the directed graph, we now compute its basic combinatorial properties. These quantities determine the architectural capacity and computational cost.

Proposition 3.5 (Directed edge count)**.** *The directed hidden edge set satisfies*

$$\left|E^{\rightarrow}_{n,m}\right| = (2n-1)\binom{m+n-2}{n}.$$

Proof. There are $2n-1$ admissible directed edge types:

$$a \to c_1, \dots, a \to c_{n-1}, \qquad c_1 \to b, \dots, c_{n-1} \to b, \qquad a \to b.$$

Consider one such type, say $a \to c_k$, corresponding to the move $\alpha \mapsto \alpha + e_{k+1} - e_0$ with $\alpha_0 \geq 1$. Setting $\alpha'_0 := \alpha_0 - 1 \geq 0$, we obtain

$$\alpha'_0 + \alpha_1 + \cdots + \alpha_n = m - 2,$$

whose number of nonnegative solutions is $\binom{m+n-2}{n}$. The same count holds for every admissible edge type, and the result follows. □

The edge count formula reveals the scaling behavior of the architecture. The factor $(2n-1)$ counts the number of directed edge types emanating from each source node, while the binomial coefficient counts how many such source nodes exist in the lattice. This combinatorial structure will also matter when comparing the depth, width, and interface size of different simplex motifs.

4. Minimal Motifs: Triangle and Tetrahedron

The smallest instances of $\mathrm{SMN}(n,m)$ occur at $m=2$, where the simplex lattice reduces to the vertices of the simplex alone. These minimal motifs make the geometry of the architecture fully transparent.

For $(n,m) = (2,2)$, the simplex is the triangle with vertices a, b, c, and the hidden nodes are precisely $\{a, c, b\}$. The input interface is the edge $F_{\text{in}} = \{a, c\}$, the output interface is $F_{\text{out}} = \{c, b\}$, and the shared face is the single point $F_{\text{mid}} = \{c\}$. The directed hidden edges are $a \to c$, $c \to b$, and $a \to b$, so the motif forms a five-stage feedforward block i, a, c, b, o with a skip path from a to b.

The next case, $(n,m) = (3,2)$, gives the tetrahedron motif with vertices a, b, c, d. The hidden nodes are $\{a, c, d, b\}$, the input interface is the face $F_{\text{in}} = \{a, c, d\}$, the output interface is $F_{\text{out}} = \{c, d, b\}$, and the shared face is the edge $F_{\text{mid}} = \{c, d\}$. The directed hidden edges are $a \to c$, $a \to d$, $c \to b$, $d \to b$, and $a \to b$, yielding a five-stage block i, a, $\{c, d\}$, b, o in which the third stage is a parallel two-node layer.

Remark 4.1 (MLP-like but not chain-structured)**.** Both motifs are MLP-like in the sense that they are feedforward, acyclic, and layered. However, they are not ordinary chain-structured MLPs. Instead, they contain geometrically induced parallel channels and skip paths, and their input and

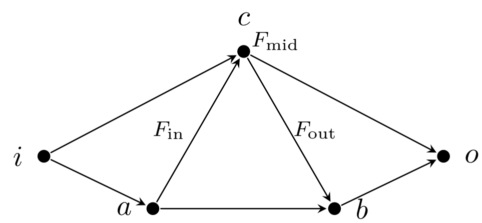

FIGURE 3. Triangle motif SMN(2, 2). Hidden nodes a, c, b with external input i and output o. Edges: $i \to a$, $i \to c$ (input injection), $a \to c$, $c \to b$ (single-step), $a \to b$ (skip), $c \to o$, $b \to o$ (readout). Input facet $F_{\text{in}} = \{a, c\}$, output facet $F_{\text{out}} = \{c, b\}$, shared interface $F_{\text{mid}} = \{c\}$.

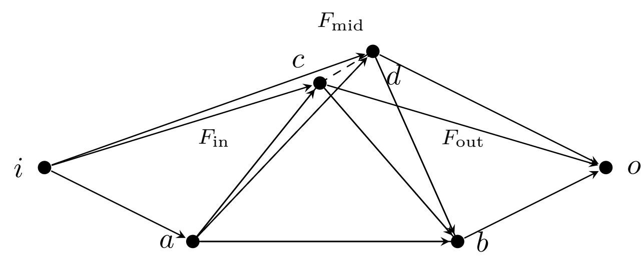

FIGURE 4. Tetrahedron motif SMN(3, 2). Hidden nodes a, c, d, b with external input i and output o. Edges: $i \to a$, $i \to c$, $i \to d$ (input injection), $a \to c$, $a \to d$, $c \to b$, $d \to b$ (single-step), $a \to b$ (skip), $c \to o$, $d \to o$, $b \to o$ (readout). Input facet $F_{\text{in}} = \{a, c, d\}$, output facet $F_{\text{out}} = \{c, d, b\}$, shared interface $F_{\text{mid}} = \{c, d\}$. Dashed edges connect to the back vertex d.

output layers are attached to entire faces rather than to a single chain of hidden layers.

5. A GEOMETRIC INTERPRETATION OF SHORT- AND LONG-TERM MEMORY

We analyze how the geometric structure of SMN(n, m) determines its long- and short-term memory properties. The reader may find it helpful to keep the tetrahedron case SMN$(3, m)$ in mind as a concrete reference throughout.

Long-term memory is encoded along the narrow backbone $\mathcal{B}_{n,m} := \{\alpha \in V_{n,m} : \alpha_2 = \cdots = \alpha_n = 0\}$, a chain of m nodes with potentials $H = 0, 2, \ldots, 2(m-1)$ from a to b. Its width is always 1 while its depth scales with m, and skip edges of type $a \to b$ allow this channel to bypass the interior entirely.

Short-term memory resides in $F_{\text{mid}} = F_{\text{in}} \cap F_{\text{out}}$, an isopotential buffer at $H = m - 1$ whose cardinality $\binom{m+n-3}{n-2}$ grows with n at fixed depth. Since $F_{\text{mid}} \subset F_{\text{out}}$, every buffer node is immediately readable at the output interface, making F_{mid} wide and shallow where the backbone is narrow and deep.

The simplex interior geometrically couples these two memory regions. Pre-midpoint backbone nodes connect into F_{mid} through edges of type $a \to c_k$, while F_{mid} connects onward to the post-midpoint backbone through edges of type $c_k \to b$. Thus the architecture is neither a pure narrow chain nor a pure broad layer. Instead, it combines a thin deep channel with a wide shared buffer, and the simplicial geometry specifies exactly how these two parts meet.

6. Open Directions

The present note provides a mathematical foundation for simplex memory networks. We outline two directions for future work: theoretical analysis of the architecture's fundamental properties, and design choices for practical deployment.

Theoretical questions concern the expressive power and continuum limits of $\text{SMN}(n, m)$ architectures. An open problem is whether simplex memory networks satisfy a universal approximation theorem as $n, m \to \infty$, in analogy with classical results for feedforward networks [4, 11]. The geometric constraints, namely fixed connectivity, potential-based orientation, and facet-attached interfaces, raise the question of whether these networks retain universal approximation capability. Another direction is to derive continuous-depth or mean-field limits, either via neural ODE formulations [3] or transport/diffusion equations on simplicial domains [6]. Such limits would provide analytical tools for studying memory consolidation, decay, and long-time behavior.

Design choices concern architectural parameters and practical deployment. The scaling parameters (n, m) control the interface-to-volume ratio $n/(n + m - 1)$: large m with small n yields deep, narrow architectures, while small m with large n produces broad, shallow networks. Systematic exploration of this design space, guided by task complexity and memory requirements, would yield practical selection guidelines.

7. Conclusion

We have introduced $\text{SMN}(n, m)$, a feedforward architecture whose hidden nodes are the barycentric lattice points of a regular n-simplex, connected by a potential-induced acyclic directed graph that routes information from the input facet F_{in} to the output facet F_{out}. The construction yields a precise geometric decomposition into a narrow deep backbone, which we identify with long-term memory, and a broad isopotential buffer F_{mid}, which we identify with short-term memory. The pair of scaling parameters (n, m)

controls the interface-to-volume ratio $n/(n+m-1)$ and thus the balance between the two memory regimes, providing a single unified handle for architectural design. We hope this framework serves as a concrete geometric foundation for further work on memory-structured neural architectures and continuum limits on simplicial domains.

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